

O9V - FASTWIND line profiles

$R = 7500$ | $v \sin i = 85 \text{ km s}^{-1}$ | $S/N = \text{none}$ | $v_{\text{turb, formal}} = 10 \text{ km s}^{-1}$

Stellar & wind parameters		
Stellar	Wind	Atmosphere
$T_{\text{eff}} = 32,882 \text{ K}$	$\dot{M} = 6.6 \times 10^{-8} M_{\odot} \text{ yr}^{-1}$	$v_{\text{turb, model}} = 10 \text{ km s}^{-1}$
$\log g = 3.92$	$v_{\infty} = 2300 \text{ km s}^{-1}$	$\xi_{\text{He}} = 2$
$R_{*} = 7.53 R_{\odot}$	$\beta = 0.8$	$f_{\text{cl, max}} = 10$
$R_{\text{max}} = 120 R_{*}$	$v_{\text{min}} = 0.1 \text{ km s}^{-1}$	$v_{\text{cl, start}}/v_{\infty} = 0.05$
$T_{\text{min}}/T_{\text{eff}} = 0.6$	$v_{\text{trans}} = 0.1 a_s$	$v_{\text{cl, max}}/v_{\infty} = 0.1$
Derived parameters		
$\log(L/L_{\odot}) = 4.776$		$M_{\text{spec}} = 17.20 M_{\odot}$
		$\log Q = -13.536$

Abundances	
Global	Individual
$Z = 1 Z_{\odot}$	$\epsilon_{\text{C}} = 8.33$
$N_{\text{He}} = 0.11$	$\epsilon_{\text{N}} = 8.46$
	$\epsilon_{\text{O}} = 8.59$

